CLASS 12 PHYSICS · CHAPTER STUDY

Rotational Dynamics

How forces turn objects, how rotation stores motion, and how conservation laws help us predict what happens next.

Maharashtra State Board preparation



EXAM PRIORITY · GUIDE

Board and MHT-CET focus by subtopic

Subtopic	Board focus	MHT-CET practice focus
Angular motion and kinematics	ω, α, θ; $\omega = \omega_0 + \alpha t$	High: equation selection
Torque and equilibrium	$\tau = rF\sin\theta$; $\Sigma F = 0$, $\Sigma\tau = 0$	High: angle and arm errors
Moment of inertia and theorems	$I = \Sigma mr^2$; parallel and perpendicular axis	High: standard bodies
Angular momentum and conservation	$L = I\omega$; zero external torque	High: ratio questions
Rolling motion	$v = R\omega$; $K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$	High: combined energy
Rotational kinetic energy	$\frac{1}{2}I\omega^2$	Medium: short numericals
Linear–angular link	$v = r\omega$; $a_t = r\alpha$	Medium

Maharashtra HSC 2026 reports list this chapter as 5 marks, or 7 marks with options. No official MHT-CET chapter-wise marks table was verified; the MHT-CET column is relative priority, not fixed marks.

ORIENTATION

What changes when an object spins?

A spin continuously changes orientation: points on one connected body can travel different distances.

By the end: describe angular motion; calculate torque and moment of inertia; apply conservation laws; solve board-style numericals; test your understanding.

THE LESSON ARC

Notice

See the motion.

Name

Choose the quantity.

Calculate

Use the condition.

Check

Test units and direction.

TWO VIEWS OF MOTION

Translation and rotation are two views of motion

Translation

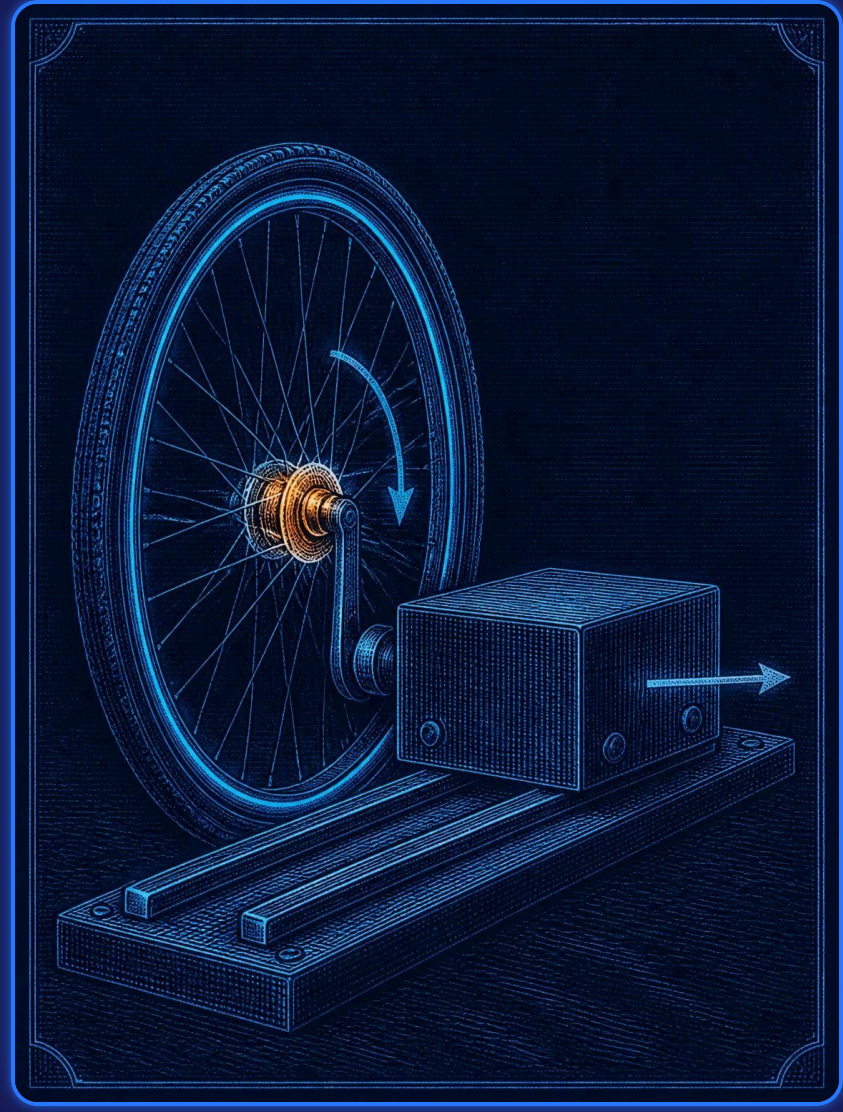
Every point moves through the same displacement.

Example: a lift moving upward.

Rotation

Points move through different distances about a fixed axis.

Examples: bicycle wheel, ceiling fan, spinning top, car wheel.



ANGULAR DISPLACEMENT

Angle is the natural language of rotation

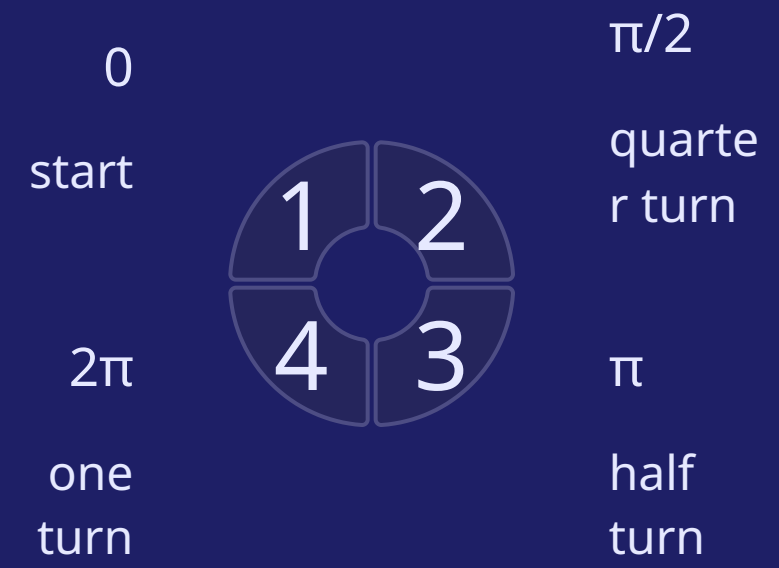
Angular displacement θ is measured in radians.

2π rad

one complete revolution

s = rθ

arc length relation



RATE OF ROTATION

Angular velocity and angular acceleration

$$\omega = \Delta\theta / \Delta t$$

angular velocity · rad s^{-1}

$$\alpha = \Delta\omega / \Delta t$$

angular acceleration · rad s^{-2}

Velocity is the rate of change of angle; acceleration is the rate of change of angular velocity.



PRACTICE CHECK

Check yourself: angular quantities

Q1. A wheel turns through 4π rad in 2 s. Find its average angular velocity.

▼ Show answer

> **Answer: 2π rad s^{-1} ≈ 6.28 rad s^{-1}.** Given Δθ = 4π rad, Δt = 2 s. Formula ω = Δθ/Δt. Substitution 4π/2. Calculation gives 2π rad s^{-1}.

Q2. Can α be zero while ω is large and non-zero?

▼ Show answer

> **Answer: Yes.** α = Δω/Δt is zero when ω stays constant, for example a steadily spinning wheel.

Short practice questions. Try first, then open the answer.

FROM ANGULAR TO LINEAR

Linear quantities follow from angular quantities

At radius r, angular motion becomes linear motion along the tangent.

$$v = r\omega$$

Tangential speed

$$a_t = r\alpha$$

Tangential acceleration

Position	Radius / speed
Near axis	small r
Midway	larger r
Rim	largest v

On the same wheel, points farther from the axis move faster.

PRACTICE · WORKED EXAMPLE 1

From revolutions to speed

Convert revolutions to radians before using angular formulas.

Given

$n = 5$ revolutions, $t = 10$ s, $r = 0.20$ m

Find

ω and rim speed v

Formula

$\theta = 2\pi n$; $\omega = \theta / t$; $v = r\omega$

Substitution

$\theta = 2\pi(5) = 10\pi$ rad; $\omega = 10\pi / 10$

Calculation

$\omega = \pi \approx 3.14$ rad s^{-1}; $v = 0.20 \times \pi \approx 0.63$ m s^{-1}

Answer

$v \approx 0.63$ m s^{-1}

Practice problem, not from a verified past paper.

CIRCULAR MOTION

Uniform circular motion needs inward acceleration

At constant speed, velocity changes direction. Acceleration points toward the centre.

$a_c = v^2/r$

or $a_c = r\omega^2$

a_t

changes speed



PRACTICE CHECK

Check yourself: circular motion

Q3. A car moves at 10 m s^{-1} on a circular track of radius 50 m. Find its centripetal acceleration.

▼ Show answer

> **Answer: 2 m s^{-2}, toward the centre.** Given v = 10 m s^{-1}, r = 50 m. Formula $a_c = v^2/r$. Substitution $10^2/50 = 100/50 = 2$ m s^{-2}.

Q4. A point 0.5 m from the axis has ω = 4 rad s^{-1}. Find its centripetal acceleration.

▼ Show answer

> **Answer: 8 m s^{-2}.** Given r = 0.5 m, ω = 4 rad s^{-1}. Formula $a_c = r\omega^2$. Substitution $0.5 \times 16 = 8$ m s^{-2}.

Short practice questions. Try first, then open the answer.

MCQ PRACTICE

MCQ practice: circular motion

Q3. At fixed ω, doubling the radius makes v:

A half · B same · C double · D zero

▼ Show answer

> **Answer: C double.** v = rω is proportional to r.

Answers sit under each question.

KINEMATICS

Rotational kinematics mirrors linear kinematics

For constant angular acceleration, replace displacement, velocity and acceleration with their angular partners.

$\omega = \omega_0 + \alpha t$

$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

$\omega^2 = \omega_0^2 + 2\alpha\theta$

$\theta = \frac{1}{2}(\omega_0 + \omega)t$

ANALOGY

Linear and rotational equations side by side

The analogy is useful only when the matching conditions apply.

Linear	Rotational	Meaning	SI unit	Check	
s	θ	displacement	m / rad		same role
v	ω	rate of change	m s^{-1} / rad s^{-1}	same role	
a	α	change of rate	m s^{-2} / rad s^{-2}	same role	
m	I	inertia	kg / kg m^2	axis matters	
F	τ	cause of change	N / N m	direction matters	

PRACTICE · WORKED EXAMPLE 2

Constant angular acceleration

Given

$\omega_0 = 4\ \text{rad s}^{-1}$, $\alpha = 2\ \text{rad s}^{-2}$, $t = 3\ \text{s}$

Find

ω and θ after 3 s

Formula

$\omega = \omega_0 + \alpha t$; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

Substitution

$\omega = 4 + 2(3)$; $\theta = 4(3) + \frac{1}{2}(2)(3)^2$

Calculation

$\omega = 10\ \text{rad s}^{-1}$; $\theta = 12 + 9 = 21\ \text{rad}$

Answer

$\omega = 10\ \text{rad s}^{-1}$, $\theta = 21\ \text{rad}$

Practice problem, not from a verified past paper.

MCQ PRACTICE

MCQ practice: angular kinematics

Q1. One revolution equals:

A π/2 rad · B π rad · C 2π rad · D 4π rad

▼ Show answer

> **Answer: C 2π rad.** One revolution is a full turn of 2π radians.

Q2. For constant ω, the angular acceleration α is:

A 0 · B 1 · C ωt · D rω

▼ Show answer

> **Answer: A 0.** α = Δω/Δt = 0 when ω does not change.

Two questions per slide. Answers sit under each question.

TURNING EFFECT

Torque measures turning effectiveness

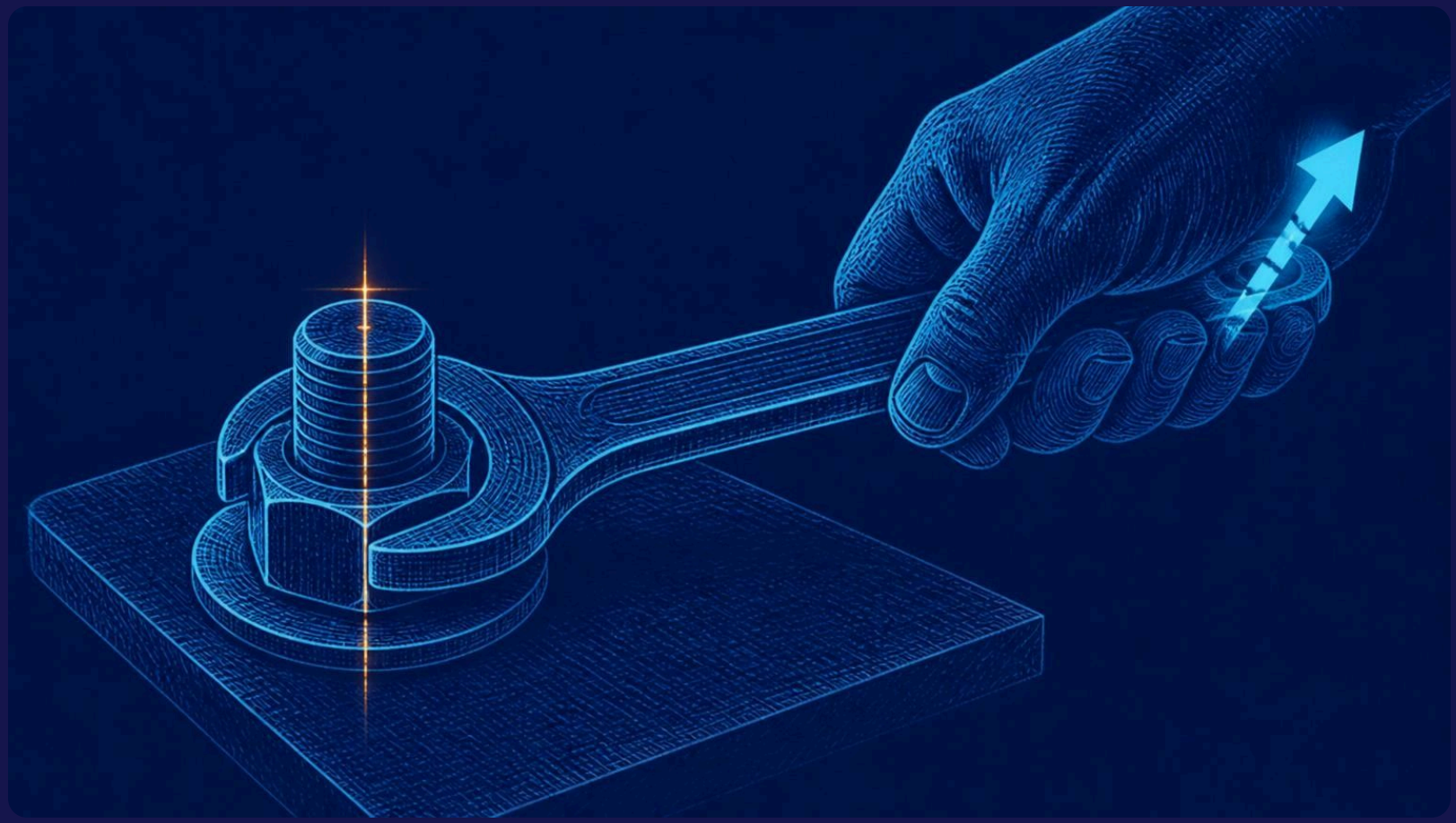
Torque is the turning effect of a force about an axis. Its size depends on force, distance, and angle.

$$\tau = rF \sin \theta$$

Magnitude of torque

N m

SI unit · direction: clockwise or anticlockwise



MOMENT ARM

Why the perpendicular distance matters

The moment arm is the perpendicular distance from the axis to the line of action of the force.

$\tau = F \times \text{perpendicular distance}$

Short handle

Small perpendicular distance → smaller torque.

Long handle

Large perpendicular distance → larger torque.

Door

Handle far from hinge.

Spanner

Longer arm turns more easily.

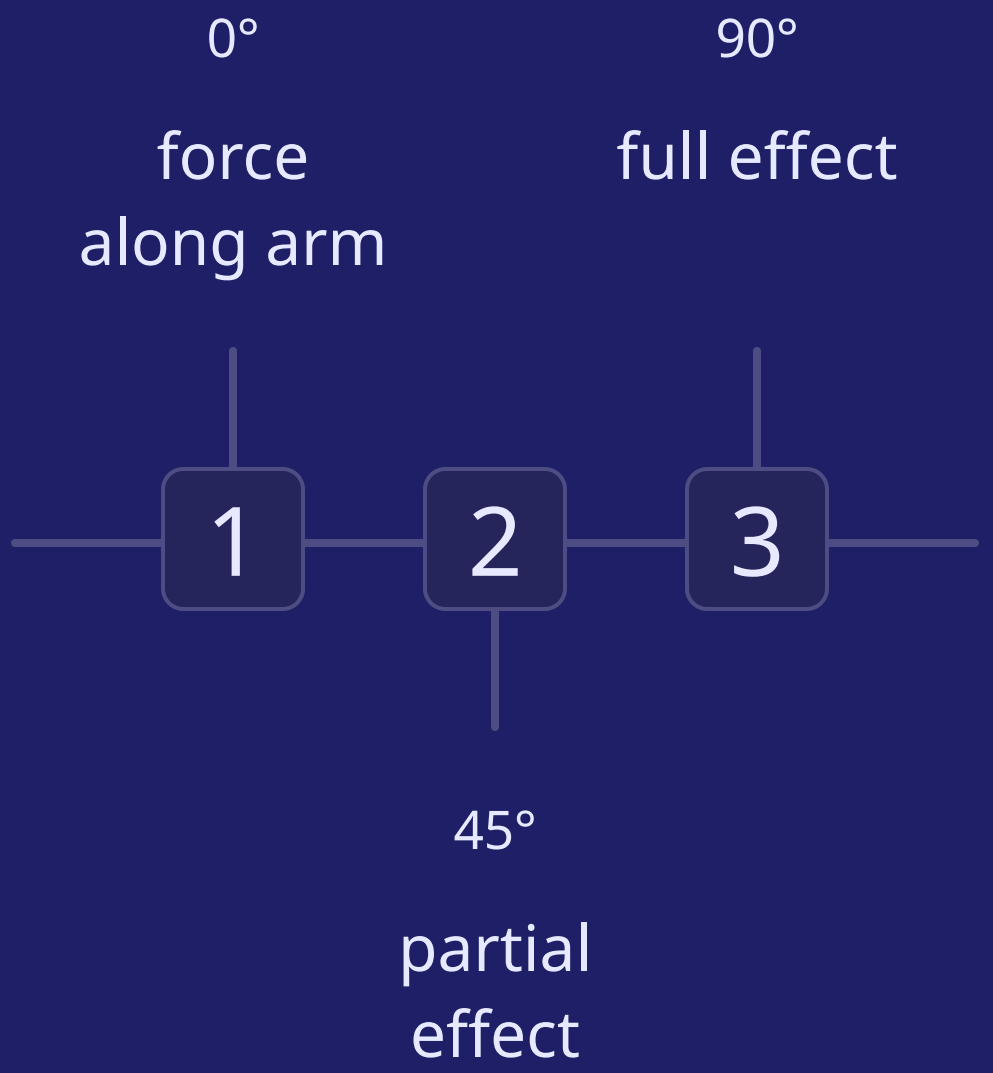
ANGLE MATTERS

Torque depends on angle

The same force produces different turning effects; only its perpendicular component contributes to torque.

90° — maximum torque

0° and 180° — zero torque when the line of action passes through the axis.



PRACTICE CHECK

Check yourself: torque

Q5. A 10 N force acts at 0.5 m from the axis, at $\theta = 90°$. Find the torque.

▼ Show answer

> **Answer: 5 N m.** Given $F = 10$ N, $r = 0.5$ m, $\theta = 90°$. Formula $\tau = rF \sin\theta$. Substitution $0.5 \times 10 \times 1$. Answer 5 N m.

Q6. The same force and distance, now at $\theta = 30°$. Find the torque.

▼ Show answer

> **Answer: 2.5 N m.** Substitution $0.5 \times 10 \times \sin 30° = 0.5 \times 10 \times 0.5 = 2.5$ N m.

Short practice questions. Try first, then open the answer.

PRACTICE · WORKED EXAMPLE 3

Torque on a spanner

Given

$F = 50\ \text{N}$, $r = 0.20\ \text{m}$, $\theta = 60°$

Find

Torque τ

Formula

$\tau = rF \sin\theta$

Substitution

$\tau = 0.20 \times 50 \times \sin 60°$

Calculation

$\tau = 10 \times 0.866 \approx 8.66\ \text{N m}$

Answer

$\tau \approx 8.66\ \text{N m}$, direction from the diagram

Practice problem, not from a verified past paper.

A COUPLE

A couple turns without translating

Two equal, opposite, parallel forces separated by a distance form a couple.

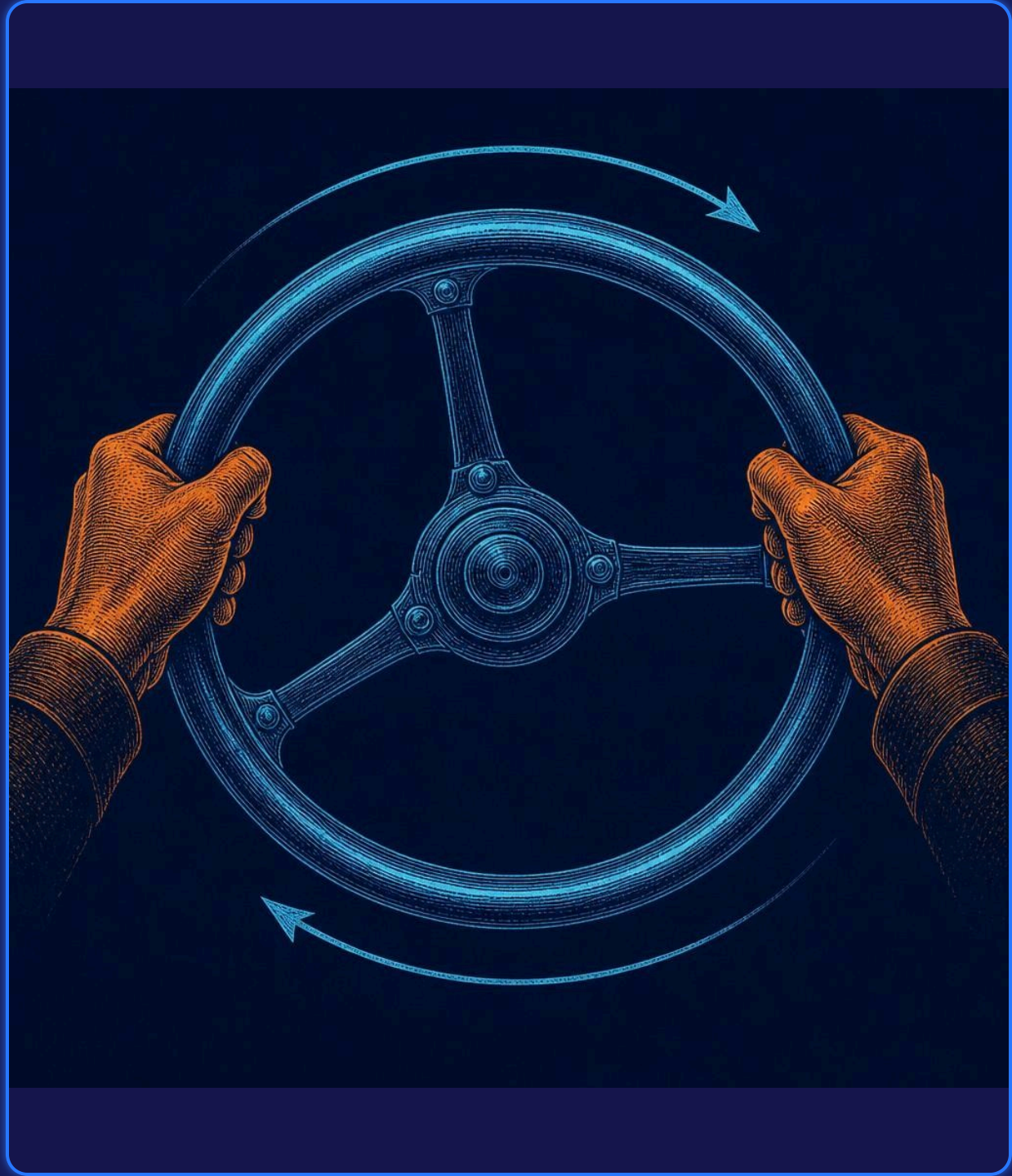
$\Sigma F = 0$

No translation

$\tau \neq 0$

Rotation remains

Moment of couple = one force × perpendicular separation.



MCQ PRACTICE

MCQ practice: torque and couples

Q6. Torque is maximum when θ is:

A 0° · B 30° · C 60° · D 90°

▼ Show answer

> **Answer: D 90°.** τ = rF sinθ is largest when sinθ = 1.

Q7. A couple has a net force of:

A zero · B maximum · C variable · D infinite

▼ Show answer

> **Answer: A zero.** Two equal, opposite, parallel forces cancel in translation.

Up to two questions per slide. Answers sit under each question.

EQUILIBRIUM

Equilibrium needs two balances

An object is fully in equilibrium only when it neither translates nor rotates.

ΣF = 0

Translational equilibrium

Στ = 0

Rotational equilibrium

A balanced beam and a stationary steering wheel satisfy both conditions.

PRACTICE · WORKED EXAMPLE 4

Balanced beam

Balance needs equal clockwise and anticlockwise torques.

Given

30 N acts 0.40 m left of the pivot; second force acts 0.60 m right of the pivot

Find

Force F that balances the beam

Formula

$\Sigma\tau = 0$: $F_1 d_1 = F_2 d_2$

Substitution

$30 \times 0.40 = F \times 0.60$

Calculation

$12 = 0.60F$, so $F = 12 / 0.60$

Answer

F = 20 N

Practice problem, not from a verified past paper.

PRACTICE CHECK

Check yourself: equilibrium

Q7. A 20 N force is 0.5 m left of a pivot. What force at 1.0 m right balances it?

▼ Show answer

> **Answer: 10 N.** Given 20 N, 0.5 m, 1.0 m. Formula $F_1 d_1 = F_2 d_2$. Substitution $20 \times 0.5 = F \times 1.0$. Answer F = 10 N.

Q8. Two equal opposite parallel forces of 5 N are 0.2 m apart. Find the couple torque and the net force.

▼ Show answer

> **Answer: τ = 1 N m; net force = 0.** Formula $\tau = F \times d$. Substitution $5 \times 0.2 = 1$ N m. Net force: $5 - 5 = 0$.

Short practice questions. Try first, then open the answer.

MCQ PRACTICE

MCQ practice: equilibrium

Q8. For equilibrium, which must hold?

A $\Sigma F = 0$ only · B $\Sigma \tau = 0$ only · C both · D neither

▼ Show answer

> **Answer: C both.** A body in full equilibrium needs both $\Sigma F = 0$ and $\Sigma \tau = 0$.

Answers sit under each question.

ROTATIONAL INERTIA

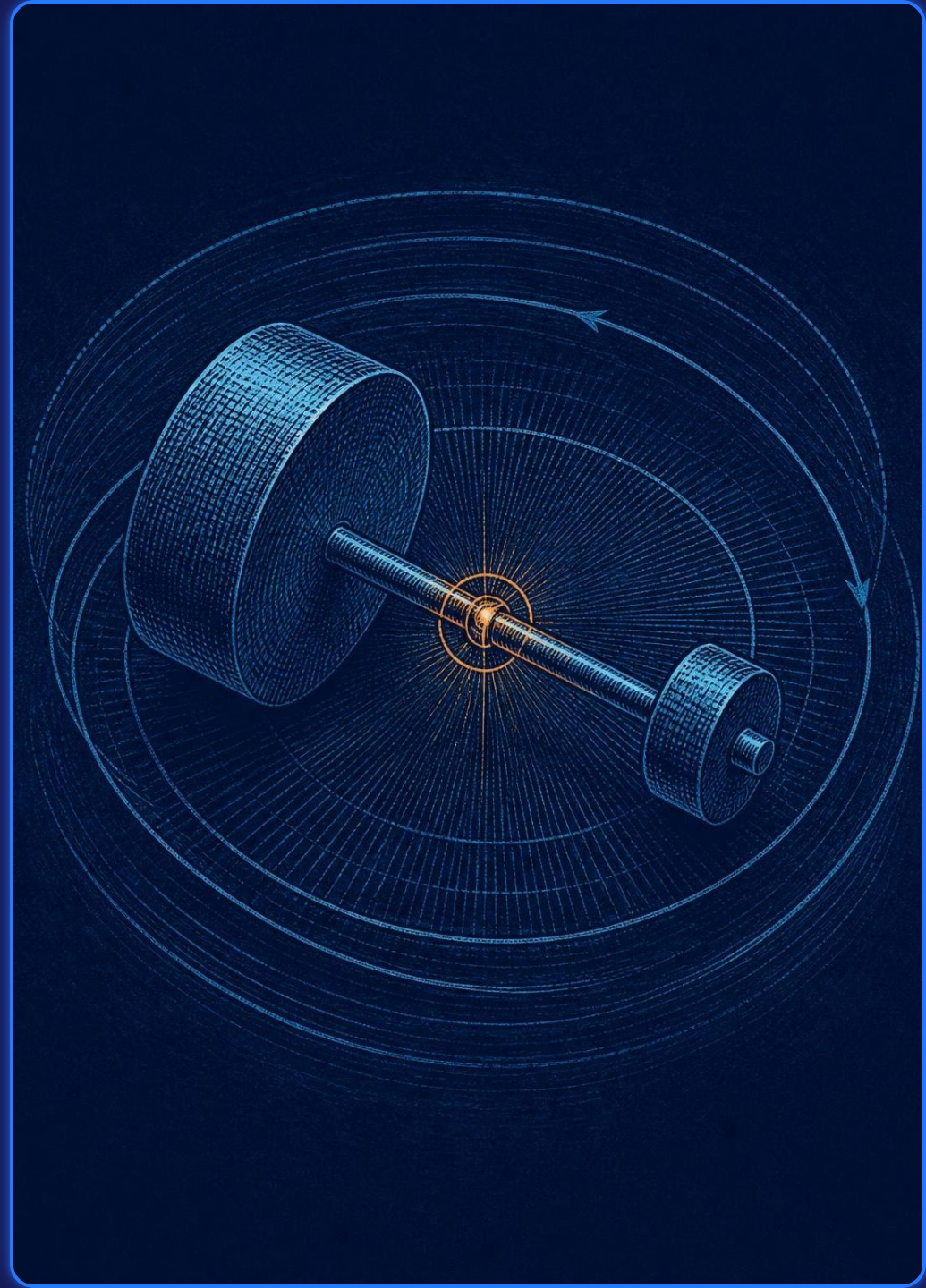
Moment of inertia is rotational inertia

Moment of inertia measures how strongly a body resists angular acceleration about a chosen axis. It depends on total mass, mass distribution, and the chosen axis.

$$I = \Sigma m_i r_i^2$$

Moment of inertia

SI unit: kg m^2 · **Dimensions:** [M L^2]



RADIUS OF GYRATION

Radius of gyration gives one useful distance

The radius of gyration k is the distance at which the whole mass could be imagined concentrated without changing the moment of inertia.

$$I = Mk^2$$

$k^2 = I/M$

Compact mass

small k → small I

Spread-out mass

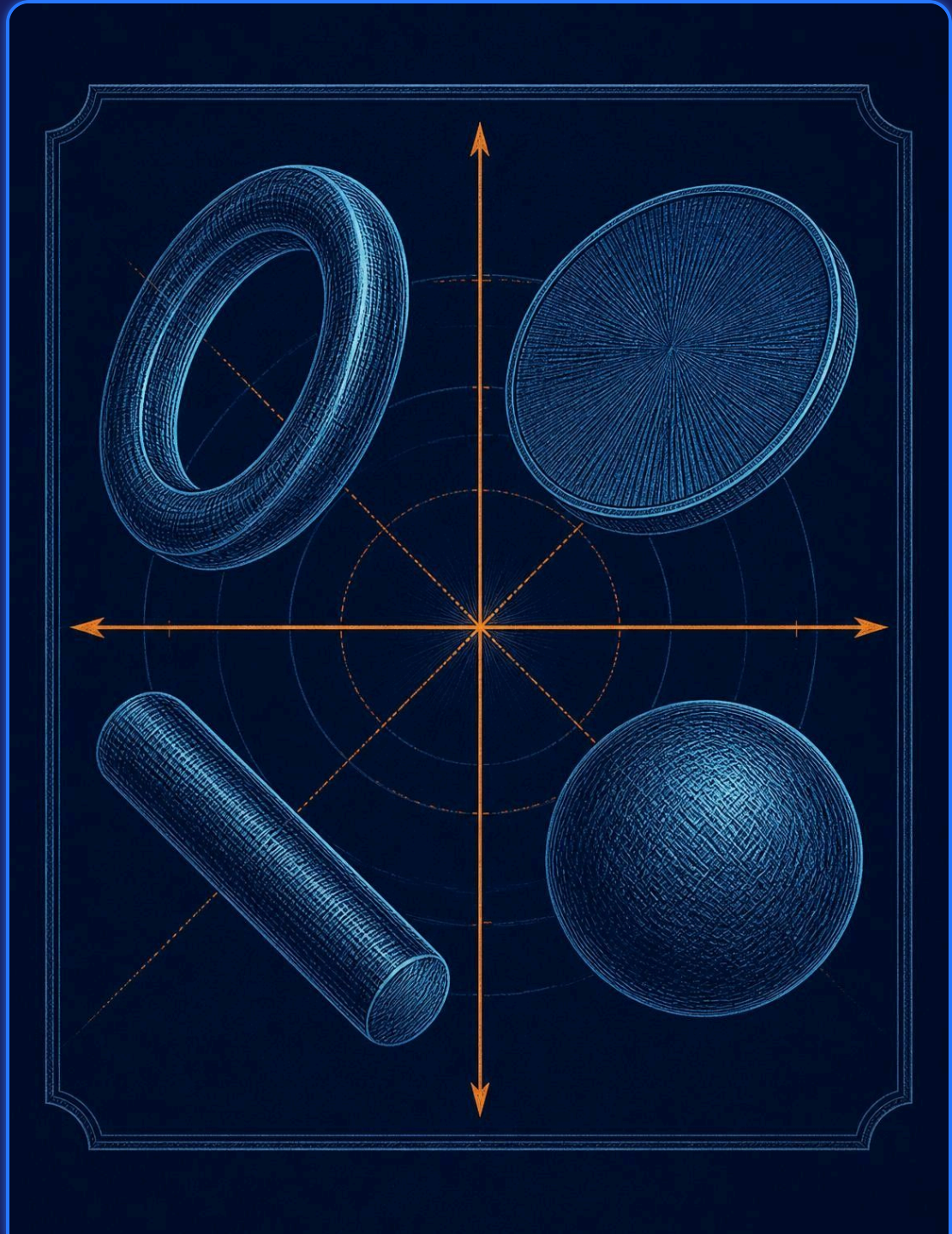
large k → large I

STANDARD BODIES

The axis changes the answer

Never choose a standard expression before identifying the body and the stated axis.

Body	Axis	Moment of inertia	Visual cue	
Thin ring	centre, perpendicular	MR^2	mass at rim	
Disc	centre, perpendicular	$\frac{1}{2}MR^2$	mass spread inward	
Rod	centre, perpendicular	$1/12\ ML^2$		axis through centre
Solid sphere	diameter	$2/5\ MR^2$	mass throughout	



PRACTICE · WORKED EXAMPLE 5

Moment of inertia of a solid disc

Dimensional check: kg × m² = kg m².

Given

M = 2 kg, R = 0.30 m; solid disc; axis through centre, perpendicular to disc

Find

I

Formula

$I = \frac{1}{2}MR^2$

Substitution

$I = \frac{1}{2}(2)(0.30)^2$

Calculation

$I = \frac{1}{2} \times 2 \times 0.09 = 0.09$ kg m²

Answer

$I = 0.09$ kg m²

Practice problem, not from a verified past paper.

SHIFTED AXES

Parallel-axis theorem

ROTATIONAL INERTIA

$$I = I_{cm} + Mh^2$$

Moving the axis away from the centre of mass increases rotational inertia: each mass element is farther from the axis.

I_{cm}

Moment of inertia about the parallel axis through the centre of mass.

h

Separation between the axes.

PLANE LAMINA

Perpendicular-axis theorem

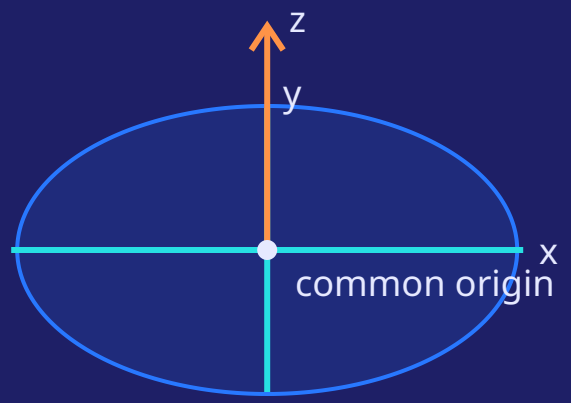
$$I_{\square} = I_x + I_y$$

For a plane lamina, x and y lie in its plane; z is perpendicular. The three axes meet at one point.

Derivation: For each mass element dm at (x, y), $r^2 = x^2 + y^2$. Summing gives $I_{\square} = \Sigma r^2 dm = \Sigma(x^2 + y^2)dm = I_x + I_y$.

Board-exam cue: state the plane-lamina condition before using the theorem.

Three axes, one origin



plane lamina (x–y plane)

PRACTICE · WORKED EXAMPLE 6

Applying the parallel-axis theorem

Given

$I_{cm} = 0.12$ kg m², $M = 3$ kg, $h = 0.20$ m; axes are parallel

Find

I about the new axis

Formula

$$I = I_{cm} + Mh^2$$

Substitution

$$I = 0.12 + 3(0.20)^2$$

Calculation

$$I = 0.12 + 0.12$$

Answer

$$\mathbf{I = 0.24\ kg\ m^2}$$

Practice problem, not from a verified past paper.

PRACTICE CHECK

Check yourself: inertia

Q9. A ring of mass 1 kg and radius 0.2 m rotates about its central axis, perpendicular to its plane. Find I.

▼ Show answer

> **Answer: 0.04 kg m².** Formula $I = MR^2$. Substitution $1 \times 0.2^2 = 0.04$ kg m².

Q10. A uniform disc of mass 1 kg and radius 0.5 m is pivoted at its rim, axis parallel to its centre axis. Find I.

▼ Show answer

> **Answer: 0.375 kg m².** Formula $I = I_{cm} + Mh^2$ with $I_{cm} = \frac{1}{2}MR^2 = 0.125$ and $h = R = 0.5$: $I = 0.125 + 1 \times 0.25 = 0.375$ kg m².

Short practice questions. Try first, then open the answer.

MCQ PRACTICE

MCQ practice: moment of inertia

Q11. Moving mass farther from the axis makes I:

A smaller · B larger · C zero · D unchanged

▼ Show answer

> **Answer: B larger.** $I = \Sigma mr^2$, so larger r gives larger I.

Q12. If $I = Mk^2$, then k is:

A force arm · B angular speed · C radius of gyration · D torque

▼ Show answer

> **Answer: C radius of gyration.** k is the radius of gyration.

Up to two questions per slide. Answers sit under each question.

MCQ PRACTICE

MCQ practice: moment of inertia of a ring

Q13. Ring, central axis perpendicular to its plane: I is:

A $\frac{1}{2}MR^2$ · B MR^2 · C $1/12\ ML^2$ · D $2/5\ MR^2$

▼ Show answer

> **Answer: B MR^2.** A thin ring has all its mass at distance R.

Answers sit under each question.

ANGULAR MOMENTUM

Angular momentum belongs to rotation

Angular momentum describes the rotational state of a body. Its direction follows the right-hand rule along the axis.

$$L = I\omega$$

rigid body about a fixed axis

$$\mathrm{kg\ m^2\ s^{-1}}$$

dimensions $[M\ L^2\ T^{-1}]$

DYNAMICAL CHAIN

Torque changes angular momentum

Torque is the rate at which angular momentum changes.

Force	Torque	Angular acceleration	Angular momentum
push or pull	turning effect	change in ω	change in L

$\tau = dL/dt$ · for constant I, $\tau = I\alpha$

PRACTICE · WORKED EXAMPLE 7

Angular momentum and torque

Given

$I = 0.50\ \text{kg m}^2$, $\omega = 12\ \text{rad s}^{-1}$; L changes by $6\ \text{kg m}^2\ \text{s}^{-1}$ in 3 s

Find

L and the average torque

Formula

$L = I\omega$; $\tau = \Delta L/\Delta t$

Substitution

$L = 0.50 \times 12$; $\tau = 6 / 3$

Calculation

$L = 6\ \text{kg m}^2\ \text{s}^{-1}$; $\tau = 2\ \text{N m}$

Answer

$L = 6\ \text{kg m}^2\ \text{s}^{-1}$, $\tau = 2\ \text{N m}$, along the axis by the right-hand rule

Practice problem, not from a verified past paper.

CONSERVATION

Conservation of angular momentum

If external torque is zero, total angular momentum remains constant.

$$I_1\omega_1 = I_2\omega_2$$

closed rotational system

I decreases

ω increases

I increases

ω decreases

Always check the external-torque condition before conserving L.

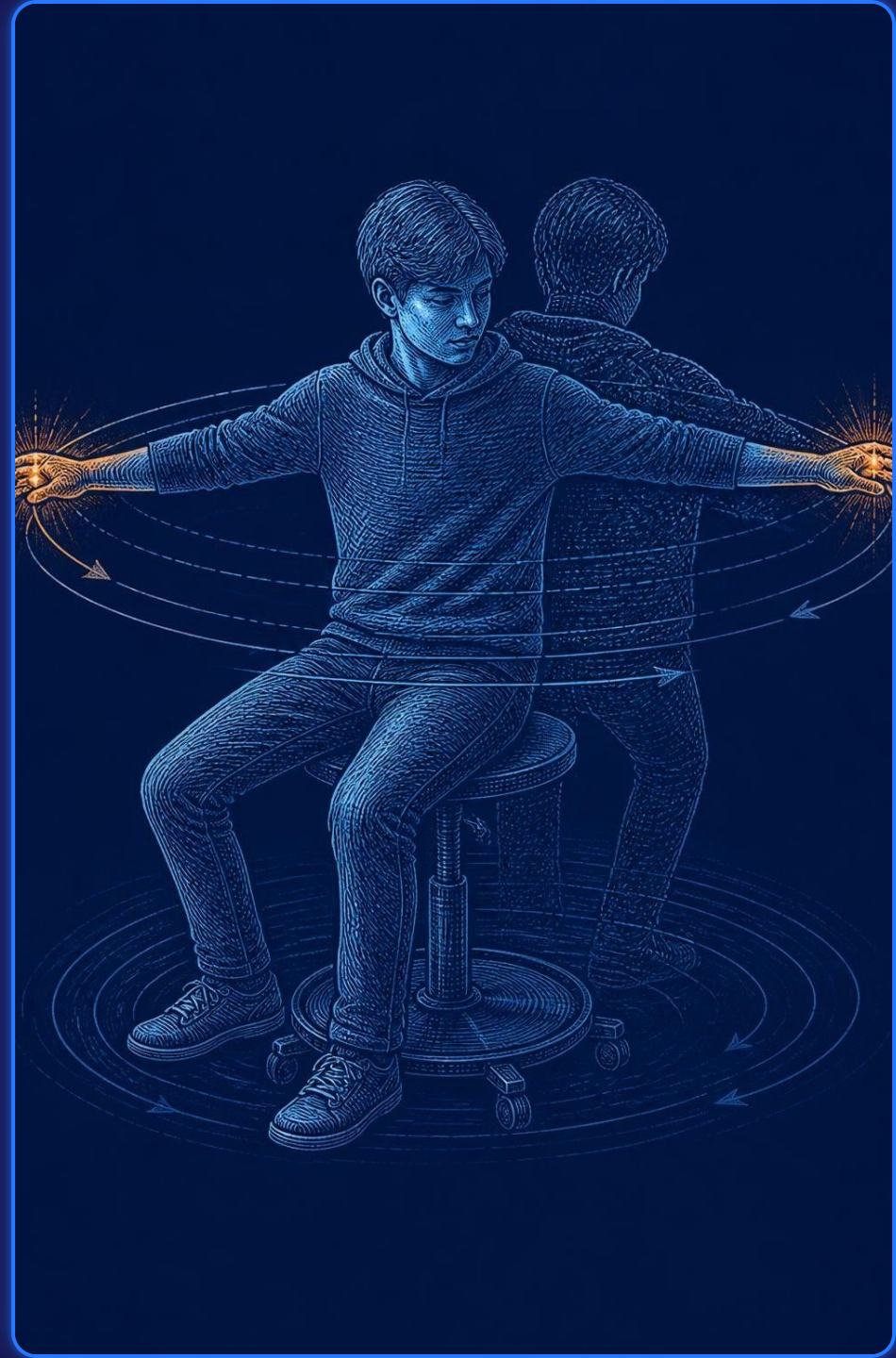
PREDICT FIRST

Skater, stool, and spinning wheel

With negligible external torque, pulling arms inward reduces I and increases angular speed.

1. **Arms out** — large I
2. **Pull inward** — redistribute mass
3. **Spin faster** — larger ω

Predict: what happens to ω when the arms move inward?



MCQ PRACTICE

MCQ practice: angular momentum

Q16. Angular momentum of a rigid body is:

A I/ω · B $I\omega$ · C $I + \omega$ · D τr

▼ Show answer

> **Answer: B $I\omega$.** $L = I\omega$ for rotation about a fixed axis.

Q17. If external torque is zero, total angular momentum is:

A doubled · B conserved · C zero · D random

▼ Show answer

> **Answer: B conserved.** L is conserved when the net external torque is zero.

Up to two questions per slide. Answers sit under each question.

PRACTICE · WORKED EXAMPLE 8

Conservation of angular momentum

Given

$I_1 = 4$ kg m^2, $\omega_1 = 3$ rad s^{-1}, $I_2 = 2$ kg m^2; no external torque

Find

ω_2

Formula

$I_1\omega_1 = I_2\omega_2$

Substitution

$4 \times 3 = 2\omega_2$

Calculation

$\omega_2 = 12 / 2 = 6$ rad s^{-1}

Answer

$\omega_2 = 6$ rad s^{-1}: I halves, so ω doubles

Practice problem, not from a verified past paper.

ENERGY

Rotational kinetic energy

A rotating body stores kinetic energy because its mass elements are moving.

$$K_r = \tfrac{1}{2} I \omega^2$$

rotational kinetic energy

$$K_t = \tfrac{1}{2} M v^2$$

translational kinetic energy

Energy rises with both moment of inertia and angular speed.

ROLLING

Pure rolling combines two motions

Translation at the centre; rotation about it. No slipping links the motions.

TRANSLATION → · SPIN ↻

$$v = R\omega$$

rolling without slipping

$$K = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I\omega^2$$

total kinetic energy



PRACTICE · WORKED EXAMPLE 9

Rolling without slipping

Given

$M = 2\ \text{kg}$, $v = 3\ \text{m s}^{-1}$, $R = 0.30\ \text{m}$, $I = 0.18\ \text{kg m}^2$; rolling without slipping

Find

ω and total kinetic energy K

Formula

$\omega = v/R$; $K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$

Substitution

$\omega = 3 / 0.30$; $K = \frac{1}{2}(2)(3)^2 + \frac{1}{2}(0.18)(10)^2$

Calculation

$\omega = 10\ \text{rad s}^{-1}$; $K = 9 + 9 = 18\ \text{J}$

Answer

$K = 18\ \text{J}$

Practice problem, not from a verified past paper.

PRACTICE CHECK

Check yourself: rolling

Q11. A disc rolls without slipping with $v = 2\ \text{m s}^{-1}$ and $R = 0.1$ m. Find ω.

▼ Show answer

> **Answer: 20 rad s^{-1}.** Formula $\omega = v/R$. Substitution $2 / 0.1 = 20\ \text{rad s}^{-1}$.

Q12. Why is the contact point momentarily at rest in pure rolling?

▼ Show answer

> **Answer: Because $v = R\omega$ makes the spin velocity cancel the centre velocity at the contact point.** Spin and translation point in opposite directions at the bottom, so their sum is zero.

Short practice questions. Try first, then open the answer.

MCQ PRACTICE

MCQ practice: rolling

Q18. Pure rolling requires:

A $v = R\omega$ · B $v = R/\omega$ · C $v = \omega/R$ · D $v = 0$

▼ Show answer

> **Answer: A $v = R\omega$.** The no-slip condition links centre speed and angular speed.

Answers sit under each question.

CHAPTER TEST

Chapter test (1 of 5)

Q1. An angle of 180° in radians is:

A π/2 · B π · C 2π · D 3π

▼ Show answer

> **Answer: B π.** 180° = π rad.

Q2. The centripetal acceleration for speed v and radius r is:

A v/r · B v^2/r · C rω · D ω/r

▼ Show answer

> **Answer: B v^2/r.** $a_c = v^2/r$ points toward the centre.

Two questions per slide. Answers sit under each question.

CHAPTER TEST

Chapter test (2 of 5)

Q3. A 20 N force acts at r = 0.3 m and θ = 30°. Torque is:

A 3 N m · B 6 N m · C 0.3 N m · D 12 N m

▼ Show answer

> **Answer: A 3 N m.** τ = 0.3 × 20 × 0.5 = 3 N m.

Q4. Torque is zero when θ equals:

A 0° · B 30° · C 60° · D 90°

▼ Show answer

> **Answer: A 0°.** sin 0° = 0; the line of action passes through the axis.

Two questions per slide. Answers sit under each question.

CHAPTER TEST

Chapter test (3 of 5)

Q5. A body in full equilibrium needs:

A $\Sigma F = 0$ only · B $\Sigma \tau = 0$ only · C both · D neither

▼ Show answer

> **Answer: C both.** Translational and rotational balance.

Q6. Moment of inertia of a uniform rod about its centre, perpendicular to its length:

A $ML^2/3$ · B $ML^2/12$ · C ML^2 · D $ML^2/2$

▼ Show answer

> **Answer: B $ML^2/12$.** Standard result for a rod about its centre.

Two questions per slide. Answers sit under each question.

CHAPTER TEST

Chapter test (4 of 5)

Q7. A solid disc, M = 2 kg, R = 0.3 m, central perpendicular axis. I is:

A 0.09 kg m² · B 0.18 kg m² · C 0.6 kg m² · D 0.045 kg m²

▼ Show answer

> **Answer: A 0.09 kg m².** $I = \frac{1}{2}MR^2 = \frac{1}{2} \times 2 \times 0.09$.

Q8. A skater pulls arms inward with no external torque. ω:

A decreases · B increases · C stays the same · D becomes zero

▼ Show answer

> **Answer: B increases.** I falls, so ω rises to keep L constant.

Two questions per slide. Answers sit under each question.

CHAPTER TEST

Chapter test (5 of 5)

Q9. Rolling without slipping with $v = 3\ \text{m s}^{-1}$ and $R = 0.3\ \text{m}$. ω is:

A $0.1\ \text{rad s}^{-1}$ · B $1\ \text{rad s}^{-1}$ · C $10\ \text{rad s}^{-1}$ · D $90\ \text{rad s}^{-1}$

▼ Show answer

> **Answer: C $10\ \text{rad s}^{-1}$.** $\omega = v/R = 3/0.3 = 10$.

Q10. Rotational kinetic energy with $I = 0.18\ \text{kg m}^2$ and $\omega = 10\ \text{rad s}^{-1}$:

A 9 J · B 18 J · C 1.8 J · D 90 J

▼ Show answer

> **Answer: A 9 J.** $\tfrac{1}{2}I\omega^2 = \tfrac{1}{2} \times 0.18 \times 100 = 9$ J.

Two questions per slide. Answers sit under each question.

EXAM PREPARATION

Board-exam preparation

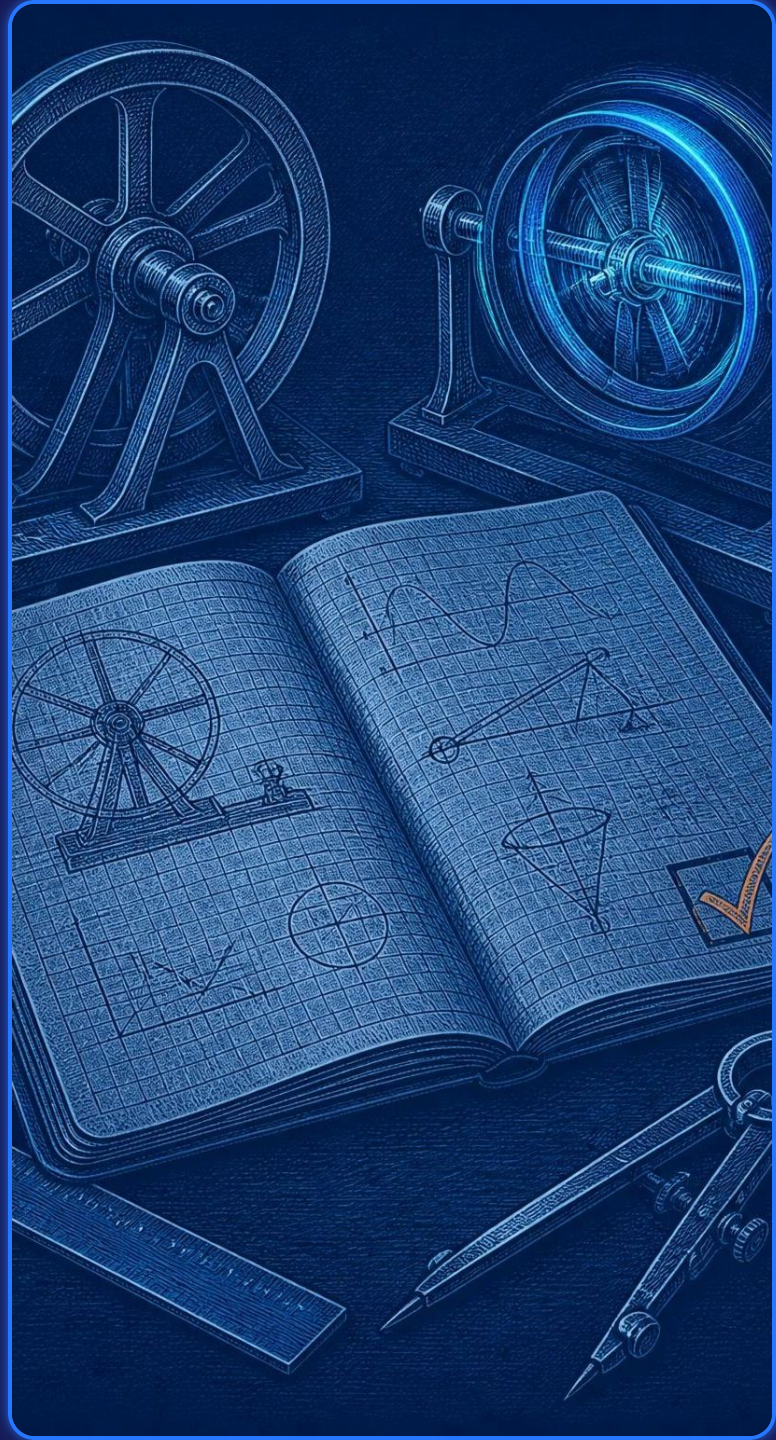
Definitions: Torque ($\tau = r F \sin\theta$); couple; moment of inertia ($I = \Sigma mr^2$); radius of gyration ($k = \sqrt{(I/M)}$); angular momentum ($L = I\omega$); pure rolling ($v = R\omega$, no slipping).

Derivations: $\omega = \omega_0 + \alpha t$ and $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$; I of a solid disc ($\frac{1}{2}MR^2$); parallel-axis theorem ($I = I_{cm} + Md^2$); $L = I\omega$ and $\tau = dL/dt$; rolling kinetic energy $\frac{1}{2}Mv^2(1 + k^2/R^2)$.

Practice: Short answers, long answers, numerical problem types, and labelled diagrams.

Full-mark answer: write the given data with units, name the principle, substitute and compute, then state the answer with unit and direction.

These are practice questions unless a previous-year source is verified.



SOLVED QUESTIONS

Subtopic-wise solved questions

Each slide is grouped by subtopic and holds at most two questions. Answers sit under each question.

Verified previous-board question means the paper year and question are confirmed from a public solution source.

Only items with a verified paper year and source are listed here.

Solved questions: moment of inertia

Q1. Verified previous-board question (2024, Maharashtra HSC Physics). Find the moment of inertia of a uniform disc of mass M and radius R about its central axis.

▼ Show answer

> **Answer: $I = \frac{1}{2} MR^2$.** For a uniform disc, the mass is spread out from the axis, and the standard result about the central perpendicular axis is $\frac{1}{2}MR^2$.

Source for Q1: OMTEX Classes, Maharashtra Board Physics 2024 solution page.

Solved questions: torque and power

Q1. Verified previous-board question (2025, Maharashtra HSC Physics). A ceiling fan rotates with a constant torque of 2 N m and angular speed of 2π rad/s. Find the power.

▼ Show answer

> **Answer: P = 4π W.** Using P = τω = (2 N m)(2π rad/s) = 4π W.

Source for Q1: OMTEX Classes, Maharashtra Board Physics 2025 solution page.

BOARD EXAM · SHORT ANSWER

Short-answer questions

Q1. Define radius of gyration and state its SI unit.

▼ Show solution

> $k = \sqrt{(I/M)}$, where I is the moment of inertia and M the mass. Its SI unit is the metre (m).

Q2. Why does a couple rotate a body without translating it?

▼ Show solution

> The two equal and opposite forces give zero net force, so there is no translation. Their torque about the centre is $F \times d$, which is not zero, so the body rotates.

BOARD EXAM · LONG ANSWER

Long-answer questions

Q1. Derive the moment of inertia of a uniform solid disc of mass M and radius R about its central axis.

▼ Show solution

> Mass per unit area $\sigma = M/(\pi R^2)$. Take a thin ring of radius r and width dr: $dm = \sigma \cdot 2\pi r\, dr$. Then $I = \int$ from 0 to R of $r^2 \cdot \sigma \cdot 2\pi r\, dr = \tfrac{1}{2}MR^2$.

Q2. State the parallel-axis theorem and use it to find the moment of inertia of a uniform rod of mass M and length L about one end.

▼ Show solution

> $I = I_{cm} + Md^2$. About the centre, $I_{cm} = ML^2/12$ and $d = L/2$, so $I_{end} = ML^2/12 + M(L/2)^2 = ML^2/3$.

BOARD EXAM · NUMERICALS

Numerical problem types

- **Angular kinematics:** find ω, α or θ using $\omega = \omega_0 + \alpha t$, $\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$ and $\omega^2 = \omega_0^2 + 2\alpha\theta$.
- **Torque:** $\tau = F \times$ perpendicular distance; find the lever arm first.
- **Equilibrium:** write $\Sigma F = 0$ and $\Sigma \tau = 0$ for a beam or ladder.
- **Moment of inertia:** choose the correct axis, then apply the parallel-axis theorem if the axis is shifted.
- **Angular momentum:** $I_1\omega_1 = I_2\omega_2$ when the external torque is zero.
- **Rolling:** $v = R\omega$; $KE = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I\omega^2$; use energy to find speed on an incline.
- **Conversions:** rpm to rad/s with $\omega = 2\pi N/60$; degrees to radians.

BOARD EXAM · DIAGRAM-BASED

Diagram-based questions

Q1. A rod pivoted at one end has a force F applied at its far end, at 60° to the rod. Draw the rod, mark the perpendicular distance from the pivot to the line of action, and find the torque about the pivot.

▼ Show solution

> The perpendicular distance is L sin60°, so τ = F L sin60°, acting to turn the rod about the pivot.

Q2. A uniform beam is pivoted at its centre. A 30 N weight hangs 1 m to the left of the pivot. How large must the balancing force be on the right if it acts 0.5 m from the pivot?

▼ Show solution

> Balance torques: 30 N × 1 m = F × 0.5 m, so F = 60 N, acting upward on the right side.

LAST-MINUTE CHECK

Last-minute revision checklist



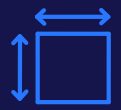
Conversions: 1 rev = 2π rad; rpm → rad/s: $\omega = 2\pi N/60$; degrees → rad: multiply by π/180.



Draw the axis and line of action before calculating torque: $\tau = rF\sin\theta$. Equilibrium: $\Sigma F = 0$ and $\Sigma\tau = 0$.



Write given data and units first. Core: $v = r\omega$; $a_t = r\alpha$; $\omega = \omega_0 + \alpha t$; $I = \Sigma mr^2$.



Axis checks: ring MR^2; disc $\frac{1}{2}MR^2$; rod about centre $ML^2/12$; rod about end $ML^2/3$; shifted axis add Md^2. Check dimensions and direction.



These are practice questions, not previous-year papers.

LAST-MINUTE CHECK

Angular momentum, rolling and units

- **Angular momentum:** $L = I\omega$; $\tau = dL/dt$. Conserved when the net external torque is zero: $I_1\omega_1 = I_2\omega_2$.
- **Rolling condition:** pure rolling means $v = R\omega$ (no slipping). Total KE = $\frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$.
- **Units and dimensions:** τ in N·m $[ML^2T^{-2}]$; I in kg·m² $[ML^2]$; L in kg·m²/s $[ML^2T^{-1}]$; ω in rad/s $[T^{-1}]$; α in rad/s² $[T^{-2}]$.

LAST-MINUTE CHECK

Common mistakes and self-test

Common mistakes

- Using degrees inside $\omega = \omega_0 + \alpha t$; convert to radians first.
- Using diameter where radius R or r is needed.
- Forgetting $\sin\theta$ when the force is not perpendicular to the lever arm.
- Using I about the wrong axis, or forgetting Md^2 for a shifted axis.
- Writing J for torque; torque is N·m.
- Applying $v = R\omega$ when the body is slipping.

Final self-test routine

1. Close notes and write the core formulas from memory.
2. Do two conversions in under a minute.
3. Solve one torque or equilibrium problem and one moment-of-inertia problem.
4. Check one rolling problem using energy.
5. Check every answer for units and dimensions.

QUICK REVISION

Formula map for quick revision

Topic	Formula	Use when	Unit / check
Angular motion	$\omega = \Delta\theta/\Delta t$; $\alpha = \Delta\omega/\Delta t$	rates are needed	rad s^{-1}, rad s^{-2}
Torque	$\tau = rF\sin\theta$	turning effect	N m; angle matters
Inertia	$I = \Sigma mr^2$; $I = Mk^2$	mass distribution	kg m^2; axis
Momentum	$L = I\omega$; $\tau = dL/dt$	rotational dynamics	kg m^2 s^{-1}
Rolling	$v = R\omega$; $K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$	no slipping	condition first

ERROR CHECK

Common formula-selection mistakes



Degrees and radians: Convert revolutions and degrees before angular formulas.



Torque arm: Use perpendicular distance, not radius.



Moment of inertia: Check the axis before choosing a formula.



Conservation: Use only when external torque is zero.



Rolling: Use $v = R\omega$ only without slipping.

FORMULA SHEET

Symbols, SI units and dimensions

Symbol	Quantity	SI unit	Dimension
θ	angular displacement	rad	1
ω	angular velocity	rad s^{-1}	T^{-1}
α	angular acceleration	rad s^{-2}	T^{-2}
r, R, h, k	radius, radius of gyration, separation	m	L
v	linear speed	m s^{-1}	L T^{-1}
F	force	N	M L T^{-2}
τ	torque	N m	M L^2 T^{-2}
M, m	mass	kg	M
I	moment of inertia	kg m^2	M L^2
L	angular momentum	kg m^2 s^{-1}	M L^2 T^{-1}
K	kinetic energy	J	M L^2 T^{-2}

Use θ in radians in every angular formula.

CONDITIONS OF USE

Use each formula only under its condition

Constant α

$\omega = \omega_0 + \alpha t$ and $\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$ only when α is constant.

Zero external torque

$I_1\omega_1 = I_2\omega_2$ only when net external torque is zero.

Parallel axes

$I = I_{cm} + Mh^2$ needs one axis through the centre of mass.

Fixed axis

$L = I\omega$ and $\tau = rF\sin\theta$ assume a clearly stated axis.

No slipping

$v = R\omega$ and $K = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I\omega^2$ only for pure rolling.

Plane lamina

$I_{\square} = I_x + I_y$ needs a flat body and perpendicular axes.

References and source note

Prescribed source: Maharashtra State Board / Balbharati Standard XII Physics textbook. Verify the exact school-issued edition and chapter details before citing them in formal work.

The publicly retrieved official PDF did not expose the Rotational Dynamics chapter in its extracted text. Therefore, textbook-specific terminology and scope should be checked against the current edition.

This deck uses the supplied project specification and standard physics relations for illustrative learning. It makes no fabricated previous-year board claims.

Check final exam wording against your current prescribed textbook.

Optional YouTube resources (not official, not endorsed):

